

A Major Project Presentation on

DineWise AI: AI-Based Restaurant Recommendation, Condition Assessment & Customer Behavior Analysis System

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Abstract

- **AI-Powered Dining Discovery:** Intelligent restaurant recommendation system tailored to urban food ecosystems.
- **Hybrid Recommendation Formulation:** Combines Content-Based TF-IDF cosine similarity (0.6) with Collaborative Filtering matrix similarity (0.4).
- **Real-Time Behavioral Telemetry:** Automatically refines user palate vectors based on weighted views, favorites, ratings, and feedback.
- **Explainable AI (XAI) Transparency:** Provides interactive score breakdowns detailing exact factor contributions (Cuisine, Zone, Budget, Diet).
- **Natural Language Intent Parser:** Extracts multi-faceted dining constraints from unstructured text with typo tolerance.
- **Offline Empirical Benchmarks:** Achieves Precision@5 of 0.600, Recall@5 of 0.875, and NDCG@5 of 0.873 on held-out evaluation sets.
- **Permanent Cloud Deployment:** Decoupled Next.js 14 frontend on Vercel and high-throughput Python FastAPI backend on Render.

Introduction

- Modern metropolitan dining ecosystems (e.g., Hyderabad) host thousands of diverse eateries across heritage, fine-dining, and street food categories.
- Consumers face cognitive choice overload, struggling to filter through scattered directories and conflicting online reviews.
- Existing platforms suffer from static popularity bias, sponsored venue saturation, and lack of genuine taste personalization.
- **DineWise AI Objective:** Automate personalized, context-aware, and explainable restaurant recommendations from user behavior.
- **Key Pipeline Stages:** User Telemetry → NLP Intent Extraction → TF-IDF Content Match → Matrix Collaborative Filtering → Hybrid Re-Ranking → Grounded XAI Output.

Literature Survey

S.No	Title	Author & Year	Description	Methodology
1	Collaborative Filtering Recommender Systems	M. D. Ekstrand et al. (2011)	Foundational analysis of user-item collaborative filtering algorithms.	Neighborhood & Matrix Factorization
2	Next Generation Recommender Systems Survey	G. Adomavicius et al. (2005)	Taxonomy of content-based, collaborative, and hybrid recommendation architectures.	Hybrid Ensemble Classification
3	Food Recommender Systems: Factors & Challenges	C. Trattner et al. (2020)	Comprehensive survey on domain-specific challenges in food & dining recommendation.	Domain-Specific Taste & Nutrition Modeling

Literature Survey (contd.)

S.No	Title	Author & Year	Description	Methodology
4	Neural Collaborative Filtering	X. He et al. (2017)	Models non-linear user-item interaction signals using deep neural networks.	Multi-Layer Perceptron + Generalized Matrix Factorization
5	Matrix Factorization Techniques for Recommenders	Y. Koren et al. (2009)	Establishes latent factor modeling over user rating matrices.	Singular Value Decomposition (SVD)

Comparison: Existing vs. Proposed System

Feature / Parameter	Existing Food Portals	DineWise AI (Proposed System)
Recommendation Engine	Distance radius & sponsored listings	Hybrid Machine Learning (0.6 Content + 0.4 Collaborative)
Explainability (XAI)	Zero explanation (Opaque black box)	Transparent modal showing exact factor percentages & rationale
Query Parsing	Rigid keyword matching	NLP constraint extraction & typo-tolerant intent parsing
Behavioral Adaptation	Static user preference registration	Real-time telemetry logging & dynamic weight recalibration
Business Intelligence	Internal ad revenue tracking	Customer behavioral clustering & live model evaluation metrics

Disadvantages of Existing Systems

- **Lack of Granular Personalization:** City-wide aggregate ratings dominate, overlooking individual dietary regimes and taste profiles.
- **Cold-Start & Long-Tail Neglect:** High-quality niche and regional dining venues remain invisible beneath heavily advertised chains.
- **Black-Box Decision Making:** Users receive no explanation of why a venue was recommended, diminishing consumer trust.
- **Rigid Keyword Search:** Search bars fail on multi-faceted queries such as 'spicy biryani under 500 in Tolichowki'.
- **Static Behavioral Modeling:** Platforms fail to adjust recommendations dynamically based on immediate user interaction signals.

Advantages of Proposed System (DineWise AI)

- **Mathematically Balanced Hybrid Fusion:** Overcomes cold-start via TF-IDF content vectors (0.6) and captures serendipity via collaborative matrix (0.4).
- **Grounded Explainability (XAI):** Provides clear user-facing rationale and quantitative feature breakdown percentages.
- **NLP Constraint & Typo Parser:** Automatically detects cuisine, Hyderabad dining zone, budget ceiling, and dietary requirements.
- **Real-Time Adaptive Telemetry:** Logs views, favorites, ratings, and recommendation feedback to continuously optimize rankings.
- **Customer Behavioral Intelligence:** Groups users into actionable dining personas (Biryani Enthusiast, Veg Explorer, Premium Diner, Budget Explorer).
- **Production Cloud Ready:** Deployed on permanent high-speed edge infrastructure with sub-50ms inference latency.

System Architecture & Dataflow

[USER INTERACTION / BROWSER]

- (Views, Searches, Favorites, Ratings, Feedback)



[PRESENTATION LAYER – NEXT.JS 14 / REACT]

- (REST API Client / Asynchronous JSON Communication)



[APPLICATION & INTELLIGENCE LAYER – FASTAPI (PYTHON)]



■ 1. NLP Search & Constraint Parser (Entity & Typo Extraction)■

■ 2. Content-Based Engine (TF-IDF Vector Cosine Similarity) ■

■ 3. Collaborative Engine (User-User Interaction Matrix) ■

■ 4. Hybrid Fusion Scorer ($0.6 * \text{Content} + 0.4 * \text{Collab}$) ■

■ 5. Explainable AI (XAI) Module (Factor Breakdown Modals) ■

6. Telemetry Logger & Behavioral Segmentation Clustering

7. Offline Evaluation Engine (Precision, Recall, NDCG)



■ (SQLAlchemy ORM / Relational Database Queries)



[DATA STORAGE LAYER – SQLITE / POSTGRESQL]

(Users, Restaurants, Cuisines, Interactions, Favorites, Ratings)

Hardware Requirements

- **Processor:** Modern Intel Core i3 / i5 / i7 or AMD Ryzen Multi-Core Processor (2.0 GHz+).
- **RAM:** Minimum 8 GB (16 GB Recommended for concurrent development & inference).
- **Storage:** Minimum 256 GB SSD (Solid State Drive) with at least 10 GB free space.
- **Display:** 1080p Full HD Display (1920 x 1080) for optimal UI visualization.
- **Network Connectivity:** Broadband Internet connection for cloud hosting and live API communication.

Software Requirements

- **Operating System:** Windows 10/11, Linux (Ubuntu 22.04 LTS), or macOS.
- **Programming Languages:** Python 3.11+, TypeScript / JavaScript (Node.js 18+).
- **Backend Framework:** FastAPI 0.110+, Uvicorn ASGI Server.
- **Frontend Framework:** Next.js 14 (App Router), React 18, Tailwind CSS, Lucide Icons.
- **Machine Learning Stack:** scikit-learn 1.4+, pandas 2.2+, NumPy 1.26+.
- **Database:** SQLite 3 (Embedded / Prototype) / PostgreSQL (Production).
- **Deployment & Tools:** Vercel (Frontend), Render (Backend), Git / GitHub, VS Code.

Offline Evaluation & Benchmark Results

Recommendation Model	Precision@5	Recall@5	F1-Score	NDCG@5	Catalog Coverage
Popularity Baseline	0.400	0.583	0.475	0.534	100.0%
Content-Based (TF-IDF)	0.700	1.000	0.824	0.981	100.0%
Collaborative Filtering	0.450	0.646	0.530	0.596	83.3%
Hybrid Engine ($\alpha=0.6$, $\beta=0.4$)	0.600	0.875	0.712	0.873	100.0%

- **Key Takeaway:** The Hybrid engine achieves an exceptional balance of high ranking quality (0.873 NDCG@5), high recall (87.5%), and complete catalog coverage (100%).

Thank You

Questions & Discussion

DineWise AI — Live Demo: <https://dinewise-ai.vercel.app>